

Enroll No.....

Class Roll No.....

II-Mid Semester Exam 2025

Branch: CSE Semester: VI

Natural Language Processing& Gen AI(CS602)

Time 1:30 Hrs

Max. Marks 20

Q.1

- a) Explain the working of **Self-Attention** in Transformer models. Why is it crucial in NLP tasks?
- b) Compare the architectures of **BERT** and **GPT** with respect to their learning objectives.

OR

- a) Describe the difference between **Pretraining** and **Fine-tuning** in LLMs. Also mention the role of **LoRA** or **PEFT** in parameter-efficient training.
- b) What are **BLEU** and **ROUGE** scores? How do they help in evaluating language models?

(CO3 7 marks)

Q.2

- a) Explain **Greedy Sampling**, **Beam Search**, and **Top-k Sampling** as techniques in text generation. Which one gives more diverse results? (CO4 7 marks)
- b) What is **Prompt Engineering**? How does it affect the output of a language model?

OR

- a) Differentiate between **Autoregressive** and **Conditional Generation** in generative AI. Provide examples.
- b) Discuss any two methods to **mitigate hallucinations or toxicity** in text generation.

Q.3

- a) What is **Cross-Modal Generative AI**? Give examples of models for **Text-to-Image** and **Text-to-Code** generation. (CO5 6 marks)
- b) Explain the concept of **Prompt Chaining** and its relevance in multi-step reasoning tasks.

OR

- a) List **two major risks** posed by generative AI (e.g., deepfakes, synthetic media). Suggest **two safety measures** adopted to control such misuse.
- b) Briefly describe the role of **EU AI Act** or **NITI Aayog Guidelines** in ensuring responsible AI development.